SCIENCE & TECHNOLOGY

How is sea level measured?

A rise of a few millimeters a year is built from tide gauges, satellites and a correction for land that moves.

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A wooden post in shallow water beside a concrete pier. · AI-generated image

The sea is never still; tides and winds move its surface daily. The oldest tool, the tide gauge, records water height against land, with the Permanent Service for Mean Sea Level collecting such records since 1933. [1] Yet land moves too, with coastal cities often sinking as groundwater is pumped out, so modern gauges are paired with satellite positioning receivers that measure the movement of the land itself. Since 1992, satellites using radar altimeters have also measured the open ocean directly by timing radar pulses. [2]

Combined, these records reveal a stark truth: according to the IPCC, global mean sea level rose by about 20 centimeters between 1901 and 2018, with the rate accelerating from 1.9 millimeters a year (1971-2006) to 3.7 millimeters (2006-2018). [3] As warming water expands and melting glaciers add water to the oceans, the fate of coasts is dictated by this relentless, millimeter-by-millimeter advance.

Sources

1. Permanent Service for Mean Sea Level (PSMSL)
https://psmsl.org/

2. NASA – Sea Level Change: observations from space
https://sealevel.nasa.gov/

3. IPCC – Sixth Assessment Report, Working Group I, Summary for Policymakers
https://www.ipcc.ch/report/ar6/wg1/chapter/summary-for-policymakers/